VARIANTS

A math slide has a seam. It just is not where the variant names are.

Every math slide after this one is at `size: portrait`, where the split fires. Eight variants render **four** structures, so there are four preprocessors — and two scaffolds that keep the whole slide, because they have nothing to slice.

What each structure does when the box gets tall

Equation and legend →

What each structure does when the box gets tall

- Equation and legend
  - The equation rides every page; the symbols page under it. A legend page without its equation is unreadable.

Step table →

What each structure does when the box gets tall (cont.)

- Step table
  - The heading opens the run, then one proof step per page — each equation back on one line at full size.

Card stack →

What each structure does when the box gets tall (cont.)

- Card stack
  - Definition, Theorem and Proof were three cards crushed into one column. Now they are three pages.

Columns →

What each structure does when the box gets tall (cont.)

- Columns
  - One labeled formulation per page, full measure, instead of two half-width columns stacked.

LINEAR REGRESSION · OLS

Equation and legend: the symbols page under the equation.

The equation, then its symbols

→

Equation and legend: the symbols page under the equation.

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

WHERE

- $\hat{\beta}$ — OLS coefficient vector

X →

Equation and legend: the symbols page under the equation. (cont.)

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

WHERE

- X — design matrix, $n \times p$

y →

Equation and legend: the symbols page under the equation. (cont.)

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

WHERE

- y — response vector, length n

Gram matrix →

Equation and legend: the symbols page under the equation. (cont.)

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

WHERE

- $X^\top X$ — Gram matrix, $p \times p$, must be invertible

The equation that was too wide for the slide, broken across lines.

The equation, then its symbols

→

The equation that was too wide for the slide, broken across lines.

$$\ell(\beta) = \sum_{i=1}^{n} \Big[y_i \log \sigma(x_i^\top \beta) \\ + (1 - y_i) \log\big(1 - \sigma(x_i^\top \beta)\big) \Big]$$

WHERE

- ℓ — log-likelihood, concave in β

the logistic link →

The equation that was too wide for the slide, broken across lines. (cont.)

$$\ell(\beta) = \sum_{i=1}^{n} \Big[y_i \log \sigma(x_i^\top \beta) + (1 - y_i) \log\big(1 - \sigma(x_i^\top \beta)\big) \Big]$$

WHERE

- σ — the logistic link

yi →

The equation that was too wide for the slide, broken across lines. (cont.)

$$\ell(\beta) = \sum_{i=1}^{n} \Big[y_i \log \sigma(x_i^\top \beta) \\ + (1 - y_i) \log\big(1 - \sigma(x_i^\top \beta)\big) \Big]$$

WHERE

- y_i — observed label, $\in \{0, 1\}$

xi →

The equation that was too wide for the slide, broken across lines. (cont.)

$$\ell(\beta) = \sum_{i=1}^{n} \Big[y_i \log \sigma(x_i^\top \beta) \\ \qquad + (1 - y_i) \log\big(1 - \sigma(x_i^\top \beta)\big) \Big]$$

WHERE

- x_i — feature vector for observation i

CHAIN RULE · FOUR STEPS

Step table: one step per page, each on a single line.

The derivation, one step per page

Step table: one step per page, each on a single line.

$f(x+h) - f(x) = f'(x)\,h + O(h^2)$	subtract $f(x)$

continues →

Step table: one step per page, each on a single line.

(cont.)

$\frac{f(x+h)-f(x)}{h} = f'(x) + O(h)$ | **divide by** $h \neq 0$

take the limit →

Step table: one step per page, each on a single line.

(cont.)

$\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = f'(x)$	take the limit

CONTINUITY · IVT

Card stack: one card per page, so a proof is read, not skimmed.

The statement, then its proof

→

Card stack: one card per page, so a proof is read, not skimmed.

DEFINITION. A function $f : [a, b] \to \mathbb{R}$ is *continuous* on $[a, b]$ if $\lim_{x \to c} f(x) = f(c)$ for every $c \in [a, b]$.

Theorem →

Card stack: one card per page, so a proof is read, not skimmed. (cont.)

THEOREM. Let f be continuous on $[a, b]$ and let y lie strictly between $f(a)$ and $f(b)$. Then there exists $c \in (a, b)$ with $f(c) = y$.

Proof →

Card stack: one card per page, so a proof is read, not skimmed. (cont.)

PROOF. Set $S = \{x \in [a, b] : f(x) < y\}$. S is non-empty and bounded; let $c = \sup S$. Continuity at c forces $f(c) = y$.

ESTIMATORS · ASYMPTOTICS

Columns: one formulation per page, at the full measure.

One formulation at a time →

Columns: one formulation per page, at the full measure.

ORDINARY LEAST SQUARES

$$\hat{\beta}_{\mathrm{OLS}} = (X^\top X)^{-1} X^\top y$$

Unbiased under exogeneity; minimum variance among linear unbiased estimators.

Ridge →

Columns: one formulation per page, at the full measure. (cont.)

RIDGE

$$\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$$

Biased, lower variance. The penalty λ buys stability when $X^\top X$ is ill-conditioned.

EFFECT SIZE · THE READING

A fixed scaffold keeps the whole slide: there is nothing to slice.

$$\hat{\beta} = \mathbf{0.42} \pm \mathbf{0.03}$$

95% CI: $[0.36,\ 0.48]$

$p < 0.001 \quad \cdot \quad n = 1{,}204$

The estimate, its uncertainty and the reading are one statement in three parts. Cut them apart and each page says less than the whole did, so this variant is left whole and rings if it ever runs long.